LESSON PLAN

AIRCRAFT ELECTRICAL EQUIPMENT, POWER GENERATION AND DISTRIBUTION (15)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	25
Total No. of Lesson Plans	35	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Introduction, effects of magnetic components on compasses
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Introduction, effects of magnetic components on compasses.
5. Specific Objectives	15.1.10.1 Trainee will be able to explain magnetic compass system.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of magnetic deviation using a magnetic compass and bar magnets; Safety: Keep magnets away from electronic devices; Supervision: Instructor monitors handling of magnets.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the magnetic compass in an aircraft?
(b) How does the Earth's magnetic field influence aircraft navigation?
(c) Why is the magnetic compass considered the primary directional reference?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Basic principles of magnetic compass operation • Earth's magnetic field and magnetic north • Compass card rotation and lubber line • Sources of magnetic interference in aircraft • Distinction between magnetic and true north	L & Ds	PPT, IB	15 min
(b)	• Hard iron magnetism and permanent deviation • Soft iron magnetism and induced deviation • Effect of aircraft structure on compass accuracy • Identification of magnetic components in airframe • Correction methods for magnetic deviation	L & Ds	PPT, PDF	20 min
(c)	• Impact of electrical systems on compass readings • Magnetic shielding techniques in aircraft design • Calibration procedures for magnetic compasses • Common errors due to magnetic components • Maintenance checks for compass accuracy	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between hard iron and soft iron magnetism in aircraft.
(b) How do magnetic components affect the accuracy of the magnetic compass?
(c) What steps are taken to correct magnetic deviation in aircraft?

12. Home Assignment and Follow Up	Prepare a brief report on how the magnetic compass system is calibrated and maintained in the next lesson plan (LP 02).

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Components of hard - iron and soft - iron magnetism.	Refer to the master syllabus reference list	LP 26

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min